1.7 Solving Equations with Variables on Both Sides

Name _____ Period ____ Date _____

Algebra I 1.7 · Student sheet

I can…

- solve equations with variables on both sides.
- identify equations that have no solution or infinitely many solutions.
- write and solve these equations for real-life problems.

Vocabulary · fill in the missing word

Variables on both sides — move the variable _____ to one side first

Identity — an _____ true for every value of the variable (infinitely many solutions)

No solution — the equation _____ to a false statement, such as 3 = 7

Part 1 · Which gym?

Gym A: $30 a month plus $5 per visit. Gym B: $10 per visit.

Visits	0	2	4	6	8
Gym A					
Gym B					

Solve 30 + 5v = 10v: _____

Which gym is better for you? Why? _____

Part 2 · Notes and examples (fill in each note, then work the example)

Variables on both sides: Collect the variable terms on _____ side.

Example 1

Solve 7x + 4 = 3x − 12.

Identity: Variables cancel and a true statement is left: _____.

No solution: Variables cancel and a false statement is left: _____.

Example 2

Solve each equation.

a. 2(x + 3) = 2x + 6 b. 3x + 5 = 3x − 1

Example 3

Taxi A charges $3 plus $2 per mile. Taxi B charges $5 plus $1.50 per mile. For how many miles do they cost the same?

Example 4

The sum of three consecutive integers is 72. Find the integers.

Part 3 · Partner practice

Partner 1

Solve $4(x - 2) = 2x + 10$.

Partner 2

Solve $6x - 2 = 2(3x - 1)$.